

Project 3

Compound Bodies - 2 bodies



4 3 constraint equations per body
4 6 constraints required

$$R_1: y = 0$$

$$x = kx? \quad \theta = 0 \quad \text{No this is wrong.}$$

$$R_2: \frac{1}{2}$$

$$M = \begin{bmatrix} m_2 & 0 & 0 \\ 0 & m_1 & 0 \\ 0 & 0 & m_1 \end{bmatrix}$$

$$\begin{bmatrix} M & C_q^T \\ C_q & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q_e \\ Q_d \end{bmatrix}$$

$$\ddot{q} = ? \rightarrow \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{y}_2 \\ \lambda \end{bmatrix}$$

Incorrect mass definition

$$F = -kx$$

$$G = -mg$$

$$Q_e = \begin{bmatrix} -kx \\ 0 \\ -mg \end{bmatrix}$$

$$Q_d = [-\dot{y} \dot{y}]$$

$$C - q \ddot{q} = \lambda$$

$$\lambda = -(\theta - g \ddot{q})$$

$$\theta_{q1} = -(x_2 - x_1)(\dot{x}_2 - \dot{x}_1) + (x_2 - x_1)(\dot{x}_2 - \dot{x}_1) + y_2 \dot{y}_2$$

$$\theta_{q2} = -\dot{y} \dot{y}$$

of course not lambda

$$M \begin{bmatrix} m_2 & 0 & 0 & -(x_2 - x_1) \\ 0 & m_1 & 0 & (x_2 - x_1) \\ 0 & 0 & m_1 & y \\ -(x_2 - x_1)(x_2 - x_1) & y & 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{y}_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -kx \\ 0 \\ -mg \\ y \cdot \ddot{y} \end{bmatrix}$$

More to
Saker
on Jabra